

# TALK2MIND: MULTIMODAL AI-BASED MENTAL WELL-BEING ASSESSMENT AND SUPPORT SYSTEM

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## Final Documentation Report

### Project Report Submitted for Internship Project Submission

#### Submitted By

**Name:** Shivam Subedar Patel

**Internship Role:** Intern (AI/ML)

**Project Title:** Talk2Mind: Multimodal AI-Based Mental Well-Being Assessment System

#### Technology Stack:

Python | Computer Vision | NLP | Scikit-Learn | TensorFlow/Keras | OpenCV | Librosa | Streamlit | Pandas | NumPy | Multimodal Late Fusion

#### Project Description:

An AI-powered mental well-being assessment system developed to analyze a person's emotional state through both visual and speech-based interactions. The project uses deep learning for facial emotion recognition, machine learning for speech emotion analysis, and a weighted late-fusion engine to generate a comprehensive Mental Well-Being Score. A Streamlit dashboard provides interactive assessments, historical tracking, and personalized recommendations.

#### Submitted To

**Company/Organization Name:** Data Vidwan

**Mentor/Manager Name:** Meet Mistry

**Date of Submission:** 30/06/2026

## 1. Project Objective

The objective of this project is to develop an AI-driven mental health decision support system that helps individuals evaluate their emotional well-being, track their mental wellness progress, and receive intelligent, personalized recommendations.

The system uses deep learning-based computer vision for facial analysis, acoustic feature modeling for speech, and dynamic risk optimization techniques to identify indicators of stress and emotional fatigue.

The project includes:

- Dataset Collection and Audio Preprocessing
- Feature extraction (MFCCs, Haarcascades)
- Facial Emotion Recognition Model (CNN)
- Speech Emotion Recognition Model (SVM)
- Mental Health Questionnaire Design
- Multimodal Late-Fusion Scoring Engine
- Personalized Recommendation logic
- Interactive Streamlit dashboard with historical tracking

## 2. Requirement Coverage

| Requirement           | Status    | Project File / Output                              |
|-----------------------|-----------|----------------------------------------------------|
| Facial Emotion Model  | Completed | models/face_emotion_model.h5 / haarcascade         |
| Speech Emotion Model  | Completed | models/speech_emotion_model.pkl                    |
| Questionnaire System  | Completed | datasets/questionnaire/mental_health_questions.csv |
| Feature Fusion Engine | Completed | utils/fusion.py                                    |
| Progress Tracking     | Completed | datasets/history.csv                               |
| Streamlit Dashboard   | Completed | dashboard/app.py                                   |

## 3. Dataset

The project utilizes multimodal datasets for training and assessment:

Facial Emotion Dataset (FER-2013 / Custom)

- Records: 30,000+ images
- Classes: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral

Speech Emotion Dataset (TESS / RAVDESS)

- Audio files in various formats
- Extracted Features: MFCCs, Pitch, Energy

## 4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Processed and standardized image and audio datasets.
- Trained a Convolutional Neural Network (CNN) for facial emotion classification.
- Extracted Mel-frequency cepstral coefficients (MFCCs) for speech analysis.
- Trained Support Vector Machine (SVM) on acoustic features.
- Implemented a standardized StandardScaler for consistent inference.
- Developed a Weighted Late-Fusion algorithm integrating vision (30%), audio (30%), and text (40%).
- Built an interactive, multipage Streamlit dashboard (Neural Diagnostics Console).
- Implemented a dynamic 'Mental Wellness Progress' tracker over time.

## 5. Methodology

### 5.1 Data Acquisition & Preprocessing

Data quality was maintained by converting all incoming audio formats (MP3, OGG, M4A, etc.) seamlessly into standardized WAV formats in-memory using FFmpeg. Face bounding boxes were extracted using OpenCV's Haar cascades prior to classification.

### 5.2 Feature Extraction

- Visual: Micro-expression mapping via CNN filters.
- Audio: 40-dimensional MFCC vectors averaged across the audio time series.
- Subjective: Weighted psychological scoring (0-4 per question, including reverse scoring).

## 6. Model Development

Multiple approaches were considered for the distinct modalities:

Facial Models Evaluated:

- MobileNetV2 (Transfer Learning)
- Custom CNN
- Haar Cascades + SVM

Speech Models Evaluated:

- Random Forest
- Support Vector Machine (SVM)
- Multi-Layer Perceptron (MLP)

## 7. Model Selection

Selected Models:

1. Facial: Custom CNN / Haar Cascade pipeline for low-latency real-time inference.
2. Speech: Support Vector Machine (SVM). Provided the best generalization on MFCC acoustic features with high accuracy and robustness to noise.

## 8. Streamlit Application

The Streamlit application is implemented in: dashboard/app.py

The dashboard contains:

Assessment Flow Page:

- Step 1: Live Camera or Image Upload for Face Emotion.
- Step 2: Real-time Audio Recording or File Upload for Speech Emotion.
- Step 3: Interactive, one-by-one self-assessment questionnaire.
- Step 4: Fused Readout generating the Wellbeing Score and Recommendations.

Dashboard & Architecture Page:

- Mental Wellness Progress (Historical Line Chart).
- Tabular history of past assessments.
- System Architecture visualizations and model metrics.

## 9. Key Findings

- Late-fusion weighting significantly outperformed single-modality evaluations.
- The inclusion of a standardized mental health questionnaire stabilizes the often highly-variant facial and acoustic predictions.
- SVM with standardized MFCCs provided faster and equally accurate results compared to heavy RNNs for the audio stream.

## 10. Recommendations

- Provide clinical resources for 'High Concern' readouts immediately.
- Encourage users to take the assessment daily to build a reliable progress chart.

## 11. Achievements

Successfully completed:

- Multimodal feature extraction (Image + Audio).
- Independent ML/DL model training.
- Implementation of a custom Decision-Level Fusion Engine.
- End-to-End Streamlit Dashboard deployment with an advanced, stylized 'Neural Diagnostics' UI.
- Real-time audio processing and dynamic history tracking.

## 12. Limitations

- Speech model relies strictly on MFCCs; adding spectrogram-based CNNs could improve environmental noise robustness.
- Recommendations are currently rule-based heuristics mapping to risk levels, rather than generated dynamically by an LLM.

## 13. Final Submission Contents

- Documentation Report (docs/Talk2Mind\_Documentation\_Report.docx)
- Machine Learning Models & Scalers
- Streamlit Dashboard (dashboard/app.py)
- Fusion & Model Utilities (utils/fusion.py, utils/model\_loader.py)
- Reports and Figures (reports/figures/)

## 14. How to Run

```
pip install -r requirements.txt
```

```
cd dashboard
```

```
streamlit run app.py
```

## 15. Final Conclusion

The Talk2Mind platform successfully delivers an end-to-end multimodal machine learning solution for mental well-being assessment.

By seamlessly integrating facial expression analysis, acoustic speech evaluation, and subjective self-reporting through a Weighted Late-Fusion engine, the system minimizes the blind spots of single-modality AI. Delivered via a highly interactive and beautifully stylized Streamlit dashboard, Talk2Mind provides users with immediate, data-driven insights and a clear way to track their mental health progress over time.